# DSA0603 Data Handling and Visualization

## Slot B

## CO1 – Assessment Test 3 (AT3)

### *Visualization Design Exercise*

**Instructions:**

For each question, write the program in the specified language (R or Python), generate the
required
visualization, and answer the interpretation sub-questions clearly with justification.

# CO1_AT3_Visualization Design Exercise

**Question 1**

**Question**

A multinational retail company has collected annual sales information from its branches across different regions. Using the given dataset, design an effective visualization that simultaneously represents **Sales, Profit, Number of Customers, Product Category, and Region**. Select suitable aesthetic mappings and apply an appropriate scale transformation wherever necessary. Justify every visualization design choice.

**R Dataset**

retail <- data.frame(

Branch=c("B1","B2","B3","B4","B5","B6","B7","B8","B9","B10",

"B11","B12","B13","B14","B15","B16","B17","B18","B19","B20"),
Region=c("North","South","East","West","North","South","East","West",
"North","South","East","West","North","South","East","West",
"North","South","East","West"),

Category=c("Electronics","Furniture","Clothing","Food","Electronics"
, "Furniture","Clothing","Food","Electronics","Furniture",
"Clothing","Food","Electronics","Furniture","Clothing",

"Food","Electronics","Furniture","Clothing","Food"),

Sales=c(450000,620000,180000,920000,350000,720000,250000,1080000
, 560000,670000,195000,980000,430000,760000,280000,1150000,
510000,690000,220000,1250000),
Profit=c(45000,95000,-12000,185000,25000,120000,18000,225000,

72000,98000,-8000,205000,51000,132000,24000,250000,

69000,115000,15000,285000),

Customers=c(1200,2500,980,4200,1800,3000,1250,5200,

2100,3300,1100,4700,1750,3600,1450,5600,

2200,3900,1600,6100)

)

**Retail Sales Visualization**

**Explanation:**

**Visualization Used: Bubble Chart**

- **X-axis:** Sales
- **Y-axis:** Profit
- **Bubble Size:** Number of Customers
- **Colour:** Product Category
- **Shape:** Region
- **Scale Transformation:** `scale_x_log10()` (Logarithmic Scale for Sales)
- **Reason for Log Scale:** Reduces skewness due to large variation in sales values and improves comparison.
- **Theme Used:** `theme_minimal()`
- **Advantages:**
  - Displays five variables in one chart.
  - Easy to compare branch performance.
  - Helps identify high-sales, high-profit, and high-customer branches.
  - Different colours and shapes clearly distinguish categories and regions.

**Code:**

library(ggplot2)

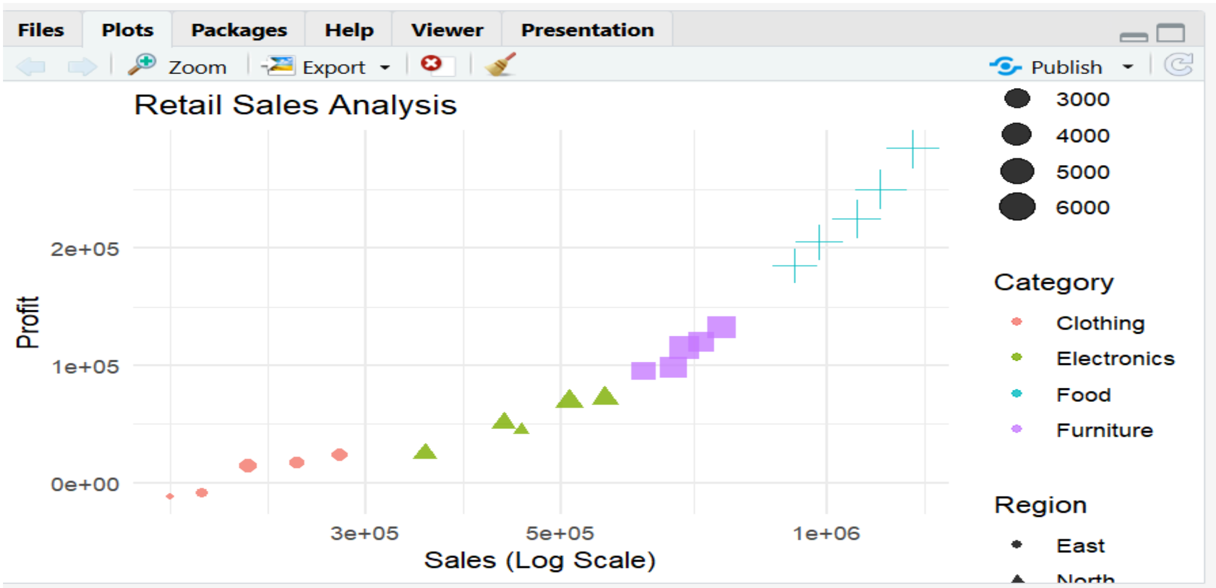
retail <- data.frame(

Branch=c("B1","B2","B3","B4","B5","B6","B7","B8","B9","B10",

```r
"B11","B12","B13","B14","B15","B16","B17","B18","B19","B20"),
Region=c("North","South","East","West","North","South","East","West",
"North","South","East","West","North","South","East","West",
"North","South","East","West"),
Category=c("Electronics","Furniture","Clothing","Food","Electronics",
"Furniture","Clothing","Food","Electronics","Furniture",
"Clothing","Food","Electronics","Furniture","Clothing",
"Food","Electronics","Furniture","Clothing","Food"),
Sales=c(450000,620000,180000,920000,350000,720000,250000,1080000,
560000,670000,195000,980000,430000,760000,280000,1150000,
510000,690000,220000,1250000),
Profit=c(45000,95000,-12000,185000,25000,120000,18000,225000,
72000,98000,-8000,205000,51000,132000,24000,250000,
69000,115000,15000,285000),
Customers=c(1200,2500,980,4200,1800,3000,1250,5200,
2100,3300,1100,4700,1750,3600,1450,5600,
2200,3900,1600,6100)
)
ggplot(retail,
      aes(x=Sales,
         y=Profit,
```

```
        size=Customers,
        color=Category,
        shape=Region))+
geom_point(alpha=0.8)+
scale_x_log10()+
labs(title="Retail Sales Analysis",
   x="Sales (Log Scale)",
   y="Profit",
   size="Customers",
   color="Category",
   shape="Region")+
theme_minimal()
```

**Output:**



**Question 2**

**Question**

The following dataset contains monthly electricity generation from different energy sources. Develop visualizations using **Cartesian Coordinates** and **Polar Coordinates**. Compare both visualizations and determine which coordinate system provides better analytical insight. Explain your design decisions.

**R Dataset**

energy <- data.frame(

Month=month.abb,

Coal=c(620,580,590,605,640,670,690,720,700,680,650,630),

Gas=c(310,295,305,320,340,355,360,370,365,350,335,320),

Solar=c(45,60,90,130,180,230,250,245,210,160,90,50),

Wind=c(80,95,120,150,170,185,195,190,175,150,110,90),

Hydro=c(140,150,160,170,180,195,205,210,200,185,165,150)

)

**(a) Cartesian Coordinate System**

**Explanation (Points)**

- **Visualization Used:** Multi-Line Chart
- **X-axis:** Month
- **Y-axis:** Electricity Generation
- **Colour:** Energy Source (Coal, Gas, Solar, Wind, Hydro)
- **Reason for Selection:**
    - Shows monthly trends clearly.
    - Easy comparison between different energy sources.

- Helps identify seasonal increases and decreases.

- **Theme Used:** `theme_minimal()`

**Code:**

```
library(ggplot2)


energy <- data.frame(
Month=month.abb,
Coal=c(620,580,590,605,640,670,690,720,700,680,650,630),
Gas=c(310,295,305,320,340,355,360,370,365,350,335,320),
Solar=c(45,60,90,130,180,230,250,245,210,160,90,50),
Wind=c(80,95,120,150,170,185,195,190,175,150,110,90),
Hydro=c(140,150,160,170,180,195,205,210,200,185,165,150)
)
energy_long <- data.frame(
Month=rep(energy$Month,5),
Source=rep(c("Coal","Gas","Solar","Wind","Hydro"),
each=nrow(energy)),
Generation=c(energy$Coal,
energy$Gas,
energy$Solar,
energy$Wind,
energy$Hydro)
```

)

ggplot(energy_long,

aes(Month,Generation,

group=Source,

color=Source))+

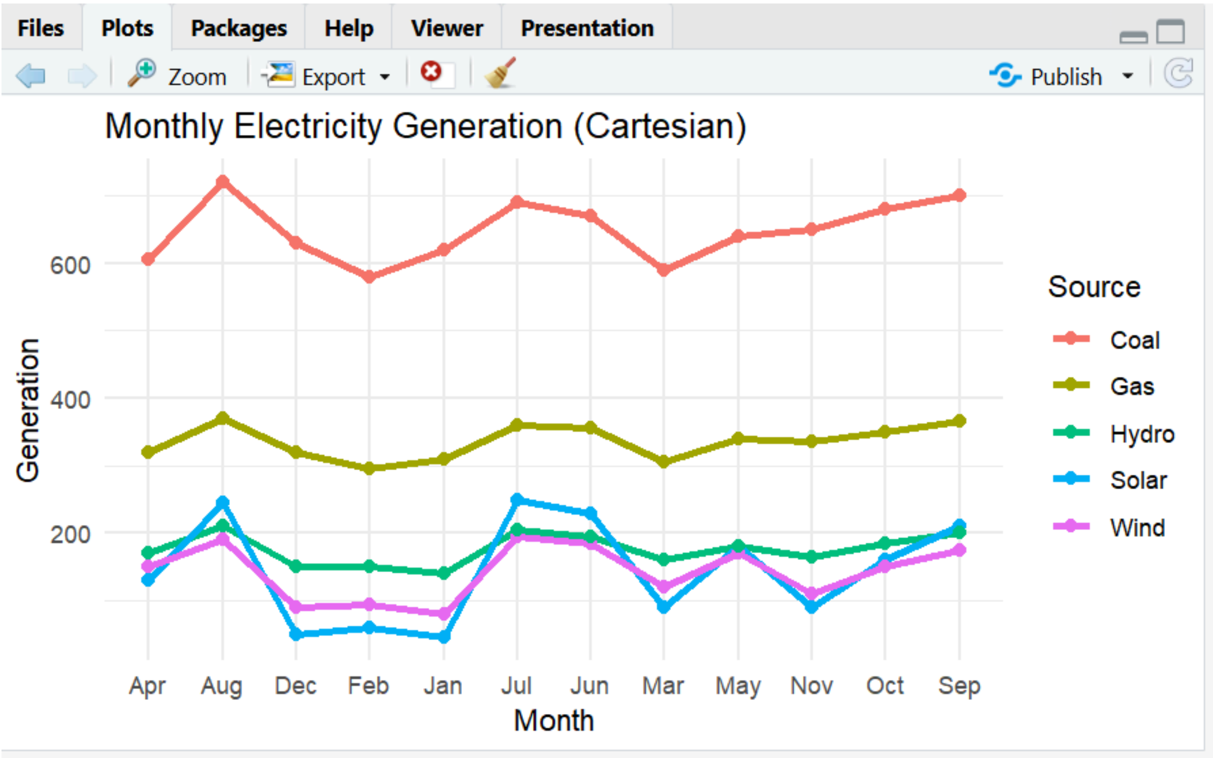
geom_line(size=1.2)+

geom_point(size=2)+

labs(title="Monthly Electricity Generation (Cartesian)")+

theme_minimal()

**Output:**




**b) Polar Coordinate System**

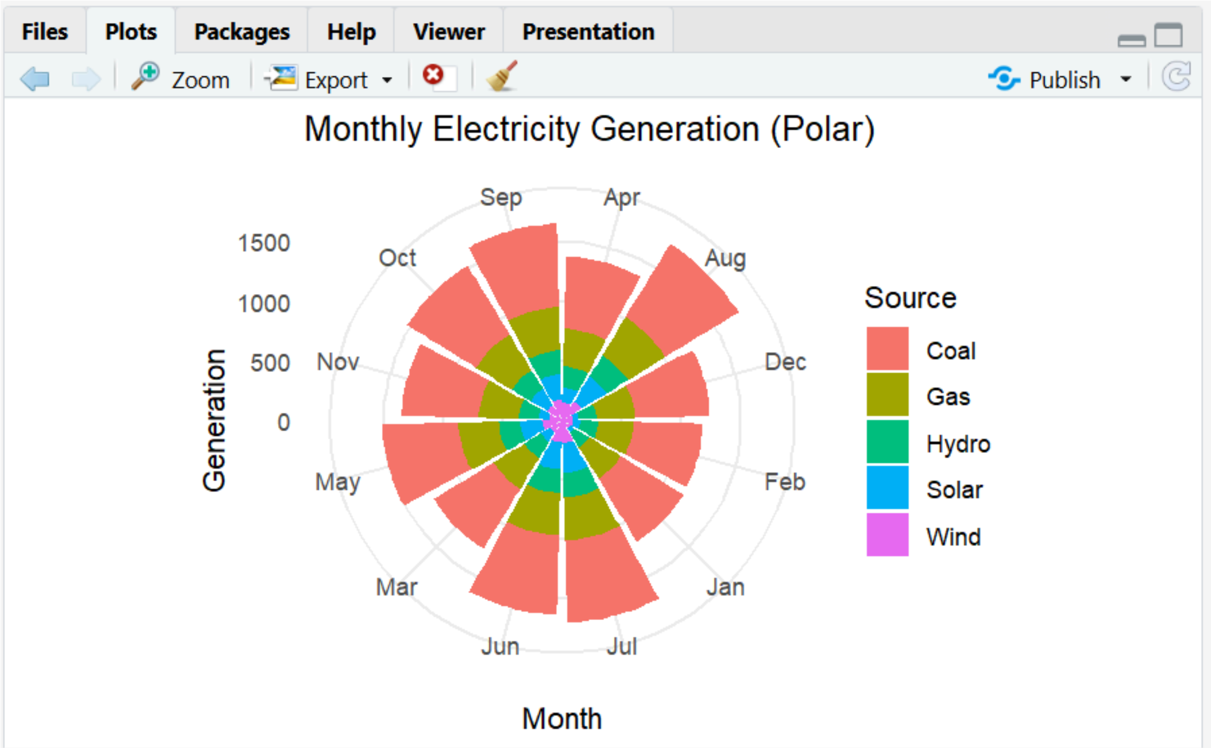
**Explanation (Points)**

- **Visualization Used:** Polar Bar Chart (`coord_polar()`)
- **Coordinate System:** Polar Coordinates
- **Colour:** Energy Source
- **Reason for Selection:**
  - Represents cyclical monthly data.
  - Attractive circular presentation.
  - Highlights seasonal patterns.
- **Theme Used:** `theme_minimal()`

**Code:**

```
ggplot(energy_long,

aes(Month,Generation,

fill=Source))+

geom_bar(stat="identity")+

coord_polar()+

labs(title="Monthly Electricity Generation (Polar)")+

theme_minimal()
```

**Output:**




**Question 3**

**Question**

An environmental monitoring agency has collected climate data from major Indian cities. Design a visualization that effectively represents **Temperature, Air Quality Index (AQI), Rainfall, and Humidity** using appropriate colour scales. Select a suitable colour palette and justify its effectiveness for accurate interpretation and accessibility.

**R Dataset**

climate <- data.frame(

City=c("Delhi","Mumbai","Chennai","Kolkata","Hyderabad",

"Bangalore","Pune","Ahmedabad","Jaipur","Lucknow",

"Bhopal","Patna","Goa","Kochi","Nagpur","Indore",

"Surat","Visakhapatnam","Coimbatore","Mysuru"), Temperature=c(44,35,38,40,39,31,33,43,45,39,

38,40,32,30,41,37,36,34,29,28),

Humidity=c(22,84,78,72,48,66,58,26,18,42,
36,45,89,91,30,34,76,82,68,64),

Rainfall=c(6,280,220,190,85,120,95,2,1,35,

48,72,360,410,12,20,180,240,130,145),

AQI=c(365,110,125,180,160,72,84,290,320,240,

175,220,58,52,270,190,145,98,60,55)

)

**Climate Data Visualization**

**Explanation (Points)**

- **Visualization Used:** Bubble Chart
- **X-axis:** Temperature
- **Y-axis:** Air Quality Index (AQI)
- **Bubble Size:** Rainfall
- **Colour:** Humidity
- **Colour Palette:** Blue to Red Gradient
- **Reason for Colour Palette:**
  - Blue indicates lower humidity.
  - Red indicates higher humidity.
  - Easy to understand and visually effective.
- **Theme Used:** `theme_minimal()`
- **Advantages:**
  - Displays four climate variables simultaneously.
  - Helps identify relationships between temperature and AQI.

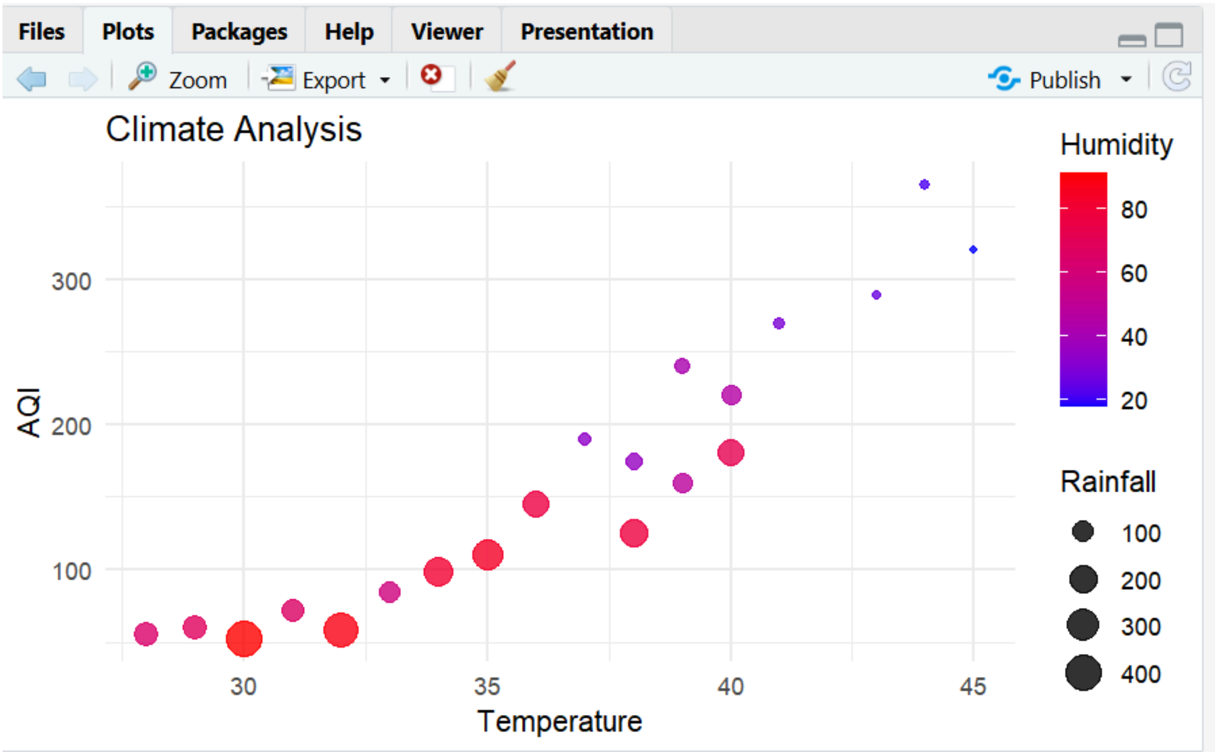
- Bubble size highlights rainfall differences.
- Colour makes humidity variation easy to observe.

**Code:**

```
library(ggplot2)

climate <- data.frame(

City=c("Delhi","Mumbai","Chennai","Kolkata","Hyderabad",

"Bangalore","Pune","Ahmedabad","Jaipur","Lucknow",

"Bhopal","Patna","Goa","Kochi","Nagpur","Indore",

"Surat","Visakhapatnam","Coimbatore","Mysuru"),

Temperature=c(44,35,38,40,39,31,33,43,45,39,

38,40,32,30,41,37,36,34,29,28),

Humidity=c(22,84,78,72,48,66,58,26,18,42,

36,45,89,91,30,34,76,82,68,64),

Rainfall=c(6,280,220,190,85,120,95,2,1,35,

48,72,360,410,12,20,180,240,130,145),

AQI=c(365,110,125,180,160,72,84,290,320,240,

175,220,58,52,270,190,145,98,60,55)

)

ggplot(climate,

aes(x=Temperature,

y=AQI,

size=Rainfall,
```

```
color=Humidity))+
geom_point(alpha=0.8)+
scale_color_gradient(low="blue",high="red")+
labs(title="Climate Analysis",
x="Temperature",
y="AQI",
size="Rainfall",
color="Humidity")+
theme_minimal()
```

**Output:**



**Question 4**

**Question**

A software company has recorded project statistics for twenty software development teams. Design a visualization that requires the use of a **nonlinear axis transformation** to represent highly skewed data. Justify the selected transformation and explain how it improves interpretation over a linear scale.

**R Dataset**

software <- data.frame(

Team=paste("T",1:20,sep=""),

LinesOfCode=c(12000,18000,25000,35000,48000,

62000,81000,96000,120000,150000,

185000,240000,310000,390000,480000,

620000,810000,1050000,1350000,1800000),

Bugs=c(35,52,60,72,88,

110,125,142,165,188,

205,240,280,320,365,

420,490,560,640,720),

Developers=c(5,7,9,10,12,

14,16,18,20,22,

25,28,30,34,38,

42,46,50,55,60),

Complexity=c(2.1,2.4,2.7,3.0,3.2,

3.5,3.8,4.1,4.5,4.8,

5.1,5.4,5.8,6.2,6.6,

7.1,7.6,8.0,8.5,9.0)

)

**Software Project Statistics**

**Explanation (Points)**

- **Visualization Used:** Scatter Plot
- **X-axis:** Lines of Code (LOC)
- **Y-axis:** Bugs
- **Bubble Size:** Number of Developers
- **Colour:** Complexity
- **Colour Gradient:** Green to Red
- **Scale Transformation:** `scale_x_log10()`
- **Reason for Log Scale:**
  - LOC values are highly skewed.
  - Compresses very large values.
  - Expands smaller values.
  - Makes trends easier to identify.
- **Theme Used:** `theme_minimal()`
- **Advantages:**
  - Better comparison of projects.
  - Reveals relationship between LOC and bugs.
  - Improves readability over a linear scale.

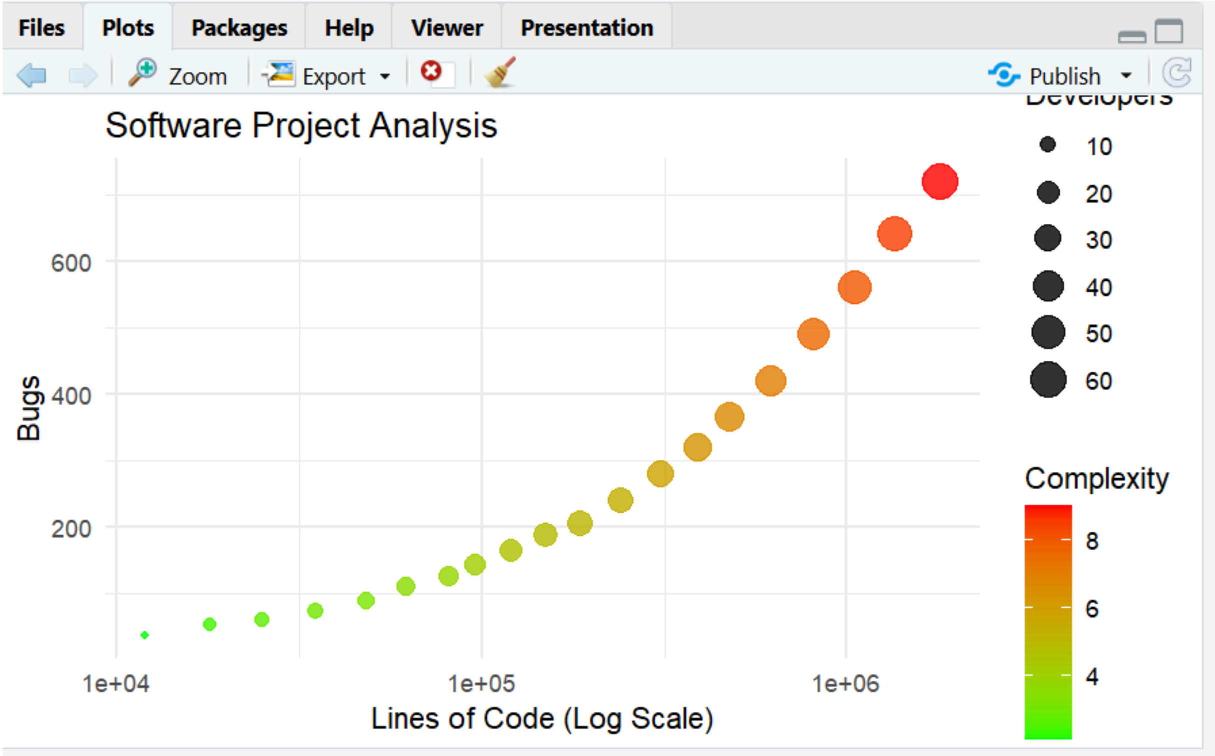
**Code:**

library(ggplot2)

```
software <- data.frame(
Team=paste("T",1:20,sep=""),
LinesOfCode=c(12000,18000,25000,35000,48000,
62000,81000,96000,120000,150000,
185000,240000,310000,390000,480000,
620000,810000,1050000,1350000,1800000),
Bugs=c(35,52,60,72,88,
110,125,142,165,188,
205,240,280,320,365,
420,490,560,640,720),
Developers=c(5,7,9,10,12,
14,16,18,20,22,
25,28,30,34,38,
42,46,50,55,60),
Complexity=c(2.1,2.4,2.7,3.0,3.2,
3.5,3.8,4.1,4.5,4.8,
5.1,5.4,5.8,6.2,6.6,
7.1,7.6,8.0,8.5,9.0)
)
ggplot(software,
aes(x=LinesOfCode,
```

```
y=Bugs,
size=Developers,
color=Complexity))+
geom_point(alpha=0.8)+
scale_x_log10()+
scale_color_gradient(low="green",high="red")+
labs(title="Software Project Analysis",
x="Lines of Code (Log Scale)",
y="Bugs",
size="Developers",
color="Complexity")+
theme_minimal()
```

**Output:**




**Question 5**

**Question**

A financial analytics company has collected performance data for twenty publicly listed companies. Develop an integrated visualization that effectively communicates **Revenue, Profit Margin, Market Capitalization, Employee Strength, and Industry Sector**. Use appropriate aesthetic mappings, coordinate systems, scales, and colour palettes to maximise interpretability while minimising perceptual bias.

**R Dataset**

```
companies <- data.frame(

Company=paste("C",1:20,sep=""),

Sector=c("IT","Banking","Healthcare","Retail","Energy",
```

```
"IT","Banking","Healthcare","Retail","Energy",
"IT","Banking","Healthcare","Retail","Energy",
"IT","Banking","Healthcare","Retail","Energy"),
Revenue=c(420,580,720,850,980,
1150,1320,1480,1650,1820,
2050,2280,2550,2840,3150,
3480,3860,4250,4680,5200),
ProfitMargin=c(8.2,15.6,11.4,9.3,18.5,
20.4,17.8,14.2,10.5,22.1,
24.5,19.8,16.7,13.4,26.2,
28.1,21.5,18.3,15.2,30.4),
Employees=c(1500,3200,2100,4200,2800,
5100,6800,3900,7200,4500,
8100,9500,6200,10300,7200,
11800,13200,8600,14500,16000),
MarketCap=c(8500,12000,9800,7600,14500,
18200,21500,16800,13200,24500,
28500,32500,24800,19200,38200,
42500,46800,35600,29800,52000) )
```

**Company Performance Visualization**
**Explanation (Points)**

- **Visualization Used:** Bubble Chart
- **X-axis:** Revenue

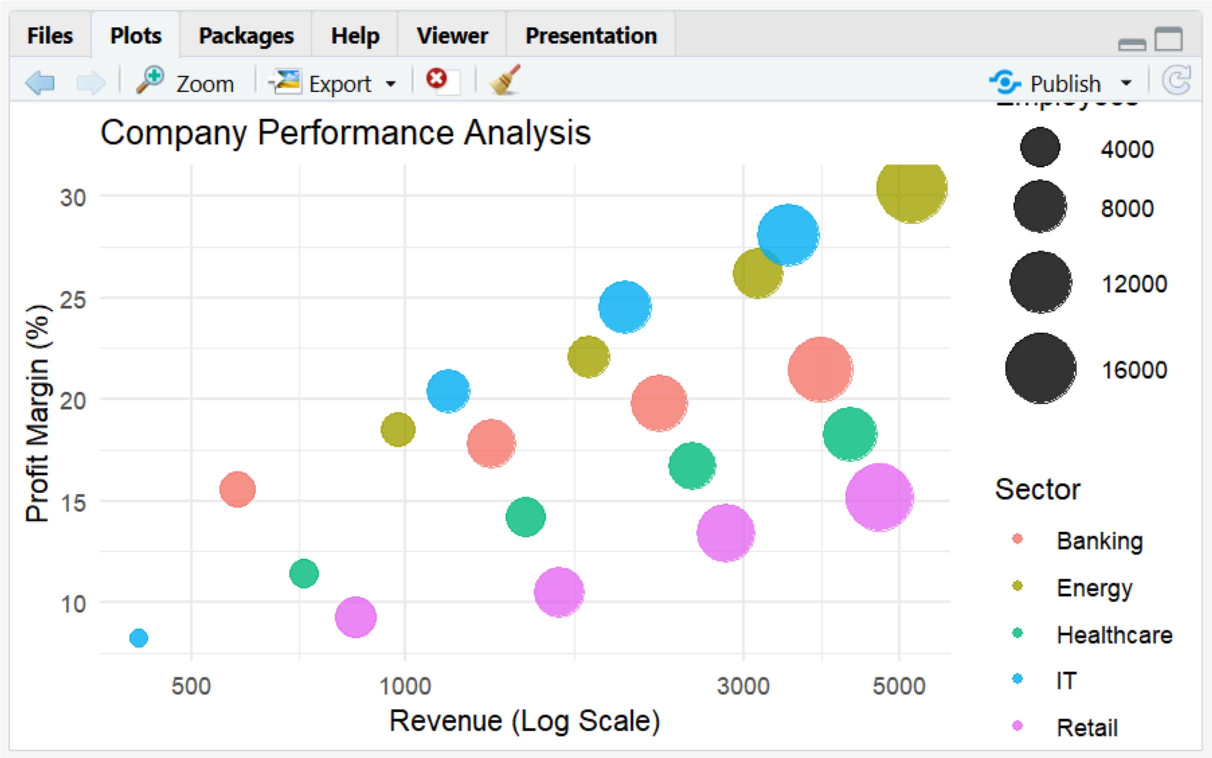
- **Y-axis:** Profit Margin
- **Bubble Size:** Employee Strength
- **Colour:** Industry Sector
- **Scale Transformation:** `scale_x_log10()` for Revenue
- **Reason for Log Scale:**
  - Revenue values vary greatly.
  - Reduces skewness.
  - Improves comparison between companies.
- **Colour Palette:** Different colours for each Industry Sector.
- **Theme Used:** `theme_minimal()`
- **Advantages:**
  - Displays five variables in one visualization.
  - Easy comparison of company performance.
  - Bubble size indicates workforce strength.
  - Colour distinguishes different industry sectors.
  - Log scale minimizes perceptual bias and improves interpretability.

**Code:**

```
library(ggplot2)
companies <- data.frame(
Company=paste("C",1:20,sep=""),
Sector=c("IT","Banking","Healthcare","Reta
il","Energy",
"IT","Banking","Healthcare","Retail","Energ
y",
"IT","Banking","Healthcare","Retail","Energ
y",
"IT","Banking","Healthcare","Retail","Energ
y"),
Revenue=c(420,580,720,850,980,
1150,1320,1480,1650,1820,
2050,2280,2550,2840,3150,
```

```
3480,3860,4250,4680,5200),
ProfitMargin=c(8.2,15.6,11.4,9.3,18.5,
20.4,17.8,14.2,10.5,22.1,
24.5,19.8,16.7,13.4,26.2,
28.1,21.5,18.3,15.2,30.4),
Employees=c(1500,3200,2100,4200,2800,
5100,6800,3900,7200,4500,
8100,9500,6200,10300,7200,
11800,13200,8600,14500,16000),
MarketCap=c(8500,12000,9800,7600,14500,
18200,21500,16800,13200,24500,
28500,32500,24800,19200,38200,
42500,46800,35600,29800,52000)
)
ggplot(companies,
aes(x=Revenue,
y=ProfitMargin,
size=Employees,
color=Sector))+
geom_point(alpha=0.8)+
scale_size_continuous(range=c(3,12))+
scale_x_log10()+
labs(title="Company Performance
Analysis",
x="Revenue (Log Scale)",
y="Profit Margin (%)",
size="Employees",
color="Sector")+
theme_minimal()
```

**Output:**


Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Company Performance Analysis
Profit Margin (%)
30
25
20
15
10
500
1000
3000
5000
Revenue (Log Scale)
4000
8000
12000
16000
Sector
Banking
Energy
Healthcare
IT
Retail